

Outline

- **What are Canonical Forms?**
- **Minterms and Maxterms**
- **Index Representation of Minterms and Maxterms**
- **Sum-of-Minterm (SOM) Representations**
- **Product-of-Maxterm (POM) Representations**
- **Representation of Complements of Functions**
- **Conversions between Representations**

Canonical Forms

- **It is useful to specify Boolean functions in a form that:**
 - **Allows comparison for equality.**
 - **Has a correspondence to the truth tables**
- **Canonical Forms in common usage:**
 - **Sum of Minterms (SOM)**
 - **Product of Maxterms (POM)**

Minterms

- **Minterms** are AND terms with every variable present in either true or complemented form.
- Given that each binary variable may appear normal (e.g., x) or complemented (e.g., $\overline{x}$), there are 2^n minterms for n variables.
- **Example:** Two variables (X and Y) produce $2 \times 2 = 4$ combinations:
 - XY (both normal)
 - $X\overline{Y}$ (X normal, Y complemented)
 - $\overline{X}Y$ (X complemented, Y normal)
 - $\overline{X}\overline{Y}$ (both complemented)
- Thus there are **four minterms** of two variables.

Maxterms

- **Maxterms** are OR terms with every variable in true or complemented form.
- Given that each binary variable may appear normal (e.g., x) or complemented (e.g., $\bar{x}$), there are 2^n maxterms for n variables.
- **Example:** Two variables (X and Y) produce $2 \times 2 = 4$ combinations:

$X + Y$ (both normal)

$X + \bar{Y}$ (x normal, y complemented)

$\bar{X} + Y$ (x complemented, y normal)

$\bar{X} + \bar{Y}$ (both complemented)

Maxterms and Minterms

- **Examples: Two variable minterms and maxterms.**

Index	Minterm	Maxterm
0	$\bar{x} \bar{y}$	$x + y$
1	$\bar{x} y$	$x + \bar{y}$
2	$x \bar{y}$	$\bar{x} + y$
3	$x y$	$\bar{x} + \bar{y}$

- The index above is important for describing which variables in the terms are true and which are complemented.

Standard Order

- Minterms and maxterms are designated with a subscript
- The subscript is a number, corresponding to a binary pattern
- The bits in the pattern represent the complemented or normal state of each variable listed in a standard order.
- All variables will be present in a minterm or maxterm and will be listed in the same order (usually alphabetically)
- Example: For variables a, b, c:
 - Maxterms: $(a + b + \bar{c})$, $(a + b + c)$
 - Terms: $(b + a + c)$, $a \bar{c} b$, and $(c + b + a)$ are NOT in standard order: - -
 - Minterms: $a b \bar{c}$, $a b c$, $\bar{a} b c$
 - Terms: $(a + c)$, $b c$, and $(a + b)$ do not contain all variables

Purpose of the Index

- The index for the minterm or maxterm, expressed as a binary number, is used to determine whether the variable is shown in the true form or complemented form.
- For Minterms:
 - “1” means the variable is “Not Complemented” and
 - “0” means the variable is “Complemented”.
- For Maxterms:
 - “0” means the variable is “Not Complemented” and
 - “1” means the variable is “Complemented”.

Index Example in Three Variables

- **Example: (for three variables)**
- **Assume the variables are called X, Y, and Z.**
- **The standard order is X, then Y, then Z.**
- **The Index 0 (base 10) = 000 (base 2) for three variables). All three variables are complemented for minterm 0 ($\bar{X}, \bar{Y}, \bar{Z}$) and no variables are complemented for Maxterm 0 (X,Y,Z).**

- **Minterm 0, called m_0 is $\bar{X}\bar{Y}\bar{Z}$.**

- **Maxterm 0, called M_0 is $(X + Y + Z)$.**
 $m_0 = \bar{X}\bar{Y}\bar{Z}$

- **Minterm 6 ? $M_6 = (X' + Y' + Z)$**

- **Maxterm 6 ?**

Minterm and Maxterm Relationship

- **Review: DeMorgan's Theorem**

$$\overline{x \cdot y} = \bar{x} + \bar{y} \text{ and } \overline{x + y} = \bar{x} \cdot \bar{y}$$

- **Two-variable example:**

$$M_2 = \bar{x} + y \text{ and } m_2 = x \cdot \bar{y}$$

Thus M_2 is the complement of m_2 and vice-versa.

- **Since DeMorgan's Theorem holds for n variables, the above holds for terms of n variables**

- **giving:** $M_i = \overline{m_i} \text{ and } m_i = \overline{M_i}$

Thus M_i is the complement of m_i .

Function Tables for Both

- **Minterms of 2 variables** **Maxterms of 2 variables**

x y	m_0	m_1	m_2	m_3
0 0	1	0	0	0
0 1	0	1	0	0
1 0	0	0	1	0
1 1	0	0	0	1

x y	M_0	M_1	M_2	M_3
0 0	0	1	1	1
0 1	1	0	1	1
1 0	1	1	0	1
1 1	1	1	1	0

- Each column in the maxterm function table is the complement of the column in the minterm function table since M_i is the complement of m_i .

Observations

- In the function tables:
 - Each minterm has one and only one 1 present in the 2^n terms (a minimum of 1s). All other entries are 0.
 - Each maxterm has one and only one 0 present in the 2^n terms. All other entries are 1 (a maximum of 1s).
- We can implement any function by "ORing" the minterms corresponding to "1" entries in the function table. These are called the minterms of the function.
- We can implement any function by "ANDing" the maxterms corresponding to "0" entries in the function table. These are called the maxterms of the function.
- This gives us two canonical forms:
 - Sum of Minterms (SOM)
 - Product of Maxterms (POM)

for stating any Boolean function.

Minterm Function Example

- **Example:** Find $F_1 = m_1 + m_4 + m_7$
- $F_1 = \bar{x} \bar{y} z + x \bar{y} \bar{z} + x y z$

x y z	inde	$m_1 + m_4 + m_7 = F_1$				
0 0 0	^x 0	0	+	0	+	0 =
0 0 1	1	1	+	0	+	0 = <u>1</u>
0 1 0	2	0	+	0	+	0 = <u>0</u>
0 1 1	3	0	+	0	+	0 = <u>0</u>
1 0 0	4	0	+	1	+	0 = <u>1</u>
1 0 1	5	0	+	0	+	0 = <u>0</u>
1 1 0	6	0	+	0	+	0 = <u>0</u>
1 1 1	7	0	+	0	+	1 = <u>1</u>

Maxterm Function Example

- Example: Implement F1 in maxterms:

$$F_1 = M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6$$

$$F_1 = (x + y + z) \cdot (\bar{x} + \bar{y} + z) \cdot (\bar{x} + y + \bar{z}) \cdot (x + \bar{y} + \bar{z}) \cdot (x + y + z)$$

x	y	z	i	M ₀	M ₁	M ₂	M ₃	M ₄	M ₅	M ₆	M ₇	F ₁
0	0	0	0	0	1	1	1	1	1	1	1	1
0	0	1	1	1	1	1	1	1	1	1	0	0
0	1	0	2	1	0	1	1	1	1	1	1	1
0	1	1	3	1	1	0	1	1	1	1	0	0
1	0	0	4	1	1	1	1	1	1	1	0	0
1	0	1	5	1	1	1	1	0	1	1	1	1
1	1	0	6	1	1	1	1	1	0	1	0	0
1	1	1	7	1	1	1	1	1	1	1	0	0

Canonical Sum of Minterms

- **Any Boolean function can be expressed as a Sum of Minterms.**
 - For the function table, the minterms used are the terms corresponding to the 1's
 - For expressions, expand all terms first to explicitly list all minterms. Do this by “ANDing” any term missing a variable v with a term $(\bar{v} + v)$.
- **Example: Implement $f = x(y + \bar{y}) + \bar{x} \bar{y}$ as a sum of minterms.**

First expand terms:

$$f = x(y + \bar{y}) + \bar{x} \bar{y}$$

Then distribute terms:

$$\text{Express as sum of minterms: } f = m_3 + m_2 + m_0$$

Another SOM Example

- **Example:** $F = A + \overline{B} C$
- **There are three variables, A, B, and C which we take to be the standard order.**
- **Expanding the terms with missing variables:**

$$\begin{aligned} F &= A (B + B') (C + C') + (A + A') B' C \\ &= ABC + ABC' + AB'C + AB'C' + AB'C + A'B'C \end{aligned}$$

- **Collect terms (removing all but one of duplicate terms):**

$$= ABC + ABC' + AB'C + AB'C' + A'B'C$$

- **Express as SOM:**

$$\begin{aligned} &= m_7 + m_6 + m_5 + m_4 + m_1 \\ &= m_1 + m_4 + m_5 + m_6 + m_7 \end{aligned}$$

Shorthand SOM Form

- From the previous example, we started with:

$$F = A + \bar{B} C$$

- We ended up with:

$$F = m_1 + m_4 + m_5 + m_6 + m_7$$

- This can be denoted in the formal shorthand:

$$F(A, B, C) = \Sigma_m(1, 4, 5, 6, 7)$$

- Note that we explicitly show the standard variables in order and drop the “m” designators.

Canonical Product of Maxterms

- Any Boolean Function can be expressed as a Product of Maxterms (POM).

- For the function table, the maxterms used are the terms corresponding to the 0's.
- For an expression, expand all terms first to explicitly list all maxterms. Do this by first applying the second distributive law, “ORing” terms missing variable v with a term equal to $V \cdot \bar{V}$ and then applying the distributive law again.

- Example: Convert to product of maxterms

$$f(x, y, z) = x + \bar{x}y$$

$$A + BC = (A + B)(A + C)$$

Apply the distributive law:

$$x + \bar{x}y = (x + \bar{x})(x + y) = 1 \cdot (x + y) = x + y$$

Add missing variable z :

$$x + y + z \cdot \bar{z} = (x + y + z)(x + y + \bar{z})$$

Express as POM: $f = M_2 \cdot M_3$

Another POM Example

- Convert to Product of Maxterms:

$$f(A, B, C) = A\bar{C} + BC + \bar{A}\bar{B}$$

- Use $x + yz = (x+y) \cdot (x+z)$ with $x = (A\bar{C} + BC)$, $y = \bar{A}$, and $z = \bar{B}$ to get:

$$f = (A\bar{C} + BC + \bar{A})(A\bar{C} + BC + \bar{B})$$

- Then use $x + \bar{x}y = x + y$ to get:

$$f = (\bar{C} + B + \bar{A})(A\bar{C} + C + \bar{B})$$

and a second time to get:

$$f = (\bar{C} + B + \bar{A})(A + C + \bar{B})$$

- Rearrange to standard order,

$$f = (\bar{A} + B + \bar{C})(A + \bar{B} + C) \text{ to give } f = M_5 \cdot M_2$$

Function Complements

- The complement of a function expressed as a sum of minterms is constructed by selecting the minterms missing in the sum-of-minterms canonical forms.
- Alternatively, the complement of a function expressed by a Sum of Minterms form is simply the Product of Maxterms with the same indices.
- Example: Given $F(x, y, z) = \Sigma_m(1, 3, 5, 7)$
 $\bar{F}(x, y, z) = \Sigma_m(0, 2, 4, 6)$
 $\bar{F}(x, y, z) = \Pi_M(1, 3, 5, 7)$

Conversion Between Forms

- To convert between sum-of-minterms and product-of-maxterms form (or vice-versa) we follow these steps:
 - Find the function complement by swapping terms in the list with terms not in the list.
 - Change from products to sums, or vice versa
- Example: Given F as before: $F(x, y, z) = \sum_m(1, 3, 5, 7)$
- Form the Complement: $F(x, y, z) = \sum_m(0, 2, 4, 6)$
- Then use the other form with the same indices – this forms the complement again, giving the other form of the original function: $F(x, y, z) = \prod_M(0, 2, 4, 6)$

Standard Forms

- **Standard Sum-of-Products (SOP) form:**
equations are written as an OR of AND terms
- **Standard Product-of-Sums (POS) form:**
equations are written as an AND of OR terms
- **Examples:**
 - SOP: $A B C + \bar{A} \bar{B} C + B$
 $(A + B) \cdot (A + \bar{B} + \bar{C}) \cdot C$
 - POS:
- **These “mixed” forms are neither SOP nor POS**
 - $A B \bar{C} + A C (A + B)$

Standard Sum-of-Products (SOP)

- **A sum of minterms form for n variables can be written down directly from a truth table.**
 - **Implementation of this form is a two-level network of gates such that:**
 - **The first level consists of n -input AND gates, and**
 - **The second level is a single OR gate (with fewer than 2^n inputs).**
- **This form often can be simplified so that the corresponding circuit is simpler.**

Standard Sum-of-Products (SOP)

- **A Simplification Example:**

- **$F(A, B, C) = \Sigma m(1, 4, 5, 6, 7)$**

- **Writing the minterm expression:**

$$F = A'B'C + AB'C' + AB'C + ABC' + ABC$$

- **Simplifying:**

$$F = A' B' C + A (B' C' + B C' + B' C + B C)$$

$$= A' B' C + A (B' + B) (C' + C)$$

$$= A' B' C + A \cdot 1 \cdot 1$$

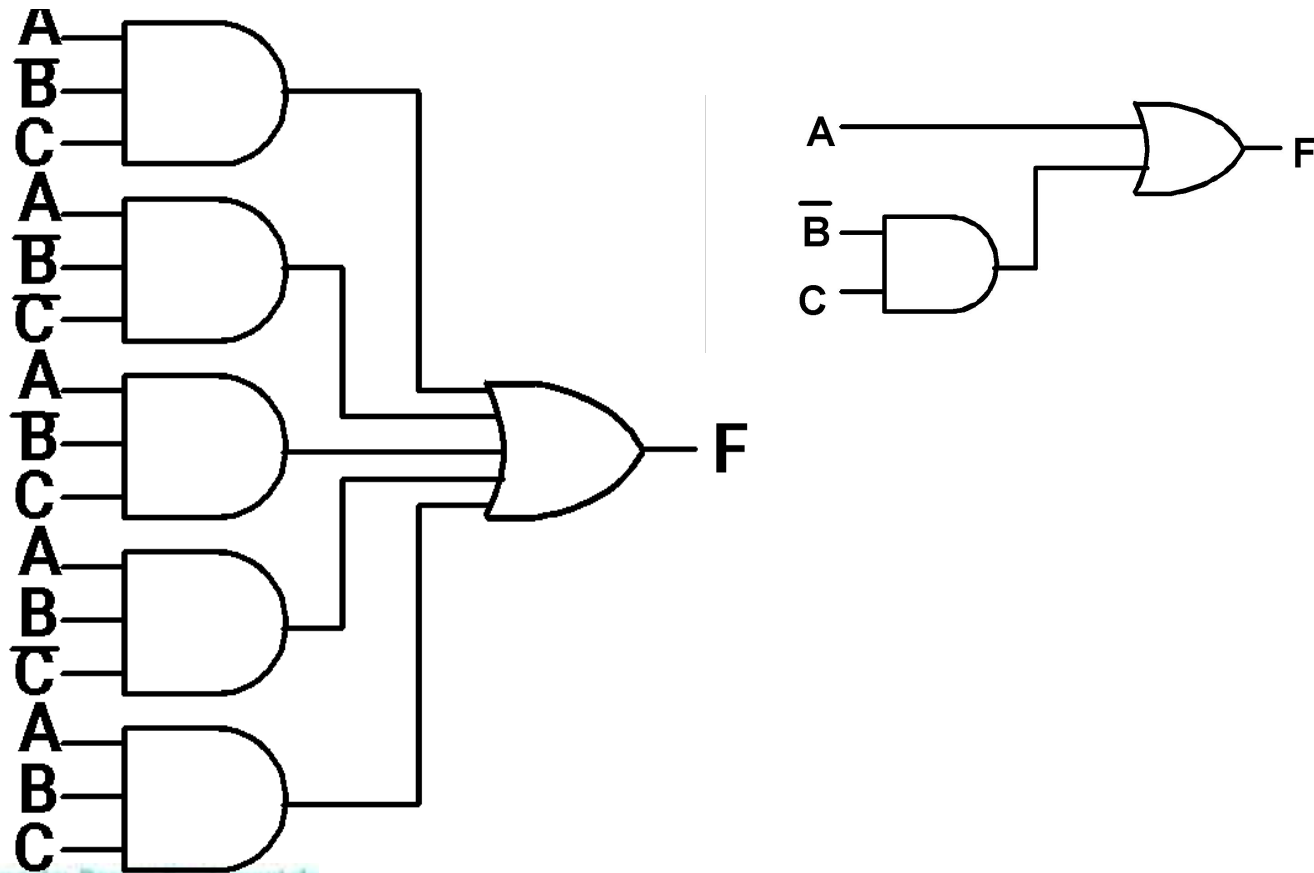
$$= A' B' C + A$$

$$= B' C + A$$

- **Simplified F contains 3 literals compared to 15 in minterm F**

AND/OR Two-level Implementation of SOP Expression

- The two implementations for F are shown below – it is quite apparent which is simpler!



SOP and POS Observations

- **The previous examples show that:**
 - **Canonical Forms (Sum-of-minterms, Product-of-Maxterms), or other standard forms (SOP, POS) differ in complexity**
 - **Boolean algebra can be used to manipulate equations into simpler forms.**
 - **Simpler equations lead to simpler two-level implementations**
- **Questions:**
 - **How can we attain a “simplest” expression?**
 - **Is there only one minimum cost circuit?**

Summary

- **What are Canonical Forms?**
- **Minterms and Maxterms**
- **Index Representation of Minterms and Maxterms**
- **Sum-of-Minterm (SOM) Representations**
- **Product-of-Maxterm (POM) Representations**
- **Representation of Complements of Functions**
- **Conversions between Representations**